

TECHNICAL MEMORANDUM

TO: Akshay Malik, Smart Cities Director, City of Philadelphia
FROM: Jinyang Xu
DATE: June 24, 2026
SUBJECT: Operationalizing Digital ROW Management Through Evidence-Based Event Workflows

1. PURPOSE

This memo evaluates Philadelphia’s Digital Right-of-Way and Mobility Improvement Project and proposes a Stage II architecture for ROWDS centered on trusted asset states and standardized event workflows.

The **Evidence-Based Event Workflow** is an operational layer built on top of ROWDS that consists of two parts: first, the **Trusted Asset-State Process (TASP)** to support asset maintenance and second, a **Standardized Event Workflow (SEW)** for cross-agency operations.

The pilot is utility work permits that requires a sidewalk closure. It is intended to test whether the workflow can simultaneously maintain trusted asset states and support cross-agency operations.

2. CURRENT STATE ASSESSMENT

The most important achievement in Stage 1 was the prototype ROWDS, which gave Philadelphia a more granular way to represent ROW assets beyond the street-centerline level. It also created a strong foundation for future service expansion and cross-agency coordination. The next challenge is whether ROWDS can operate inside daily workflows and continue to reflect real-world changes.

First, ROWDS asset states still need a sustainable trust mechanism.

Stage 1 data generation relied heavily on vendor extraction and City QA/QC. That approach was appropriate when building up the ROWDS, but it is not enough for long-term operations and maintenance. While Cityworks has been identified as a potential asset management data source, relying on a single source does not constitute a mature workflow. ROW conditions can also change through events like permitting, utility work, and field inspections/sensors. Without a multi-source evidence process, ROWDS may still rely too heavily on one system to interpret a complex and changing ROW environment.

Second, ROWDS provides a shared language, but not yet a shared workflow.

Stage 1 demonstrated that permit data can be converted into ROWDS-compliant Event Policy objects. That process, however, remained a prototype. The Events API was designed and tested, but it was not connected to a live data feed. The permit intake workflow was not yet deployed as a running service connected to the LaneClosure Master API. Without a standardized event workflow, interagency coordination will still depend on manual map checks, email confirmation, and conversations outside the system.

We should move ROWDS from “shared data to view” toward “shared process to operate.”

3. PROPOSED STRATEGY

In Stage II, an **Evidence-Based Event Workflow** should be built to respond directly to the two gaps identified above: asset trust and workflow standardization.

First, ROWDS needs a **Trusted Asset-State Process (TASP)**.

ROWDS should not update asset states from any single source automatically. TASP first needs to set up a list of relatively trusted info sources that will be taken into consideration as asset-change signals, such as Cityworks, permits, utility work, field inspections, sensors.

Asset-change signals should first be standardized into ROWDS-compatible asset-state events. Each event should include a standardized information framework, including source system, source record ID, timestamp, geometry, affected ROWDS asset ID, change type, proposed state, confidence level, and review status. This creates a consistent schema for comparing heterogeneous signals before they modify the ROWDS asset layer.

A State Resolution Framework can then compare signals across sources and determine whether to keep the existing record, create a temporary state, propose a permanent asset update, or route the case to manual review. This can begin as a rules-based process inspired by Bayesian evidence accumulation, with different confidence thresholds for different asset types and risk levels.

Second, a **Standardized Event Workflow (SEW)** for cross-agency operations.

Street closures, utility work, curb regulation changes, and transit disruptions should be converted into trackable ROWDS events. Each event should identify affected ROWDS assets and route the case to the responsible agencies. Each agency's review status and decision should be stored in the same event record rather than scattered across emails or off-system conversations. The cooperation process should be discussed and coordinated, but a standard event work-flow must be at the ROWDS level.

In this Evidence-Based Event Workflow, TASP ensures the reliability of the underlying asset data, while SEW makes that data actionable through daily interagency operations. The two layers are tightly integrated rather than operating independently. TASP maintains the trusted asset foundation that supports SEW activities, while validated SEW events feed back into TASP as evidence for maintaining and updating ROWDS asset states.

4. PILOT PROPOSAL: Utility Work Permits that Requires a Sidewalk Closure

This use case was selected because it involves multiple agencies, temporary changes to ROW assets, and high coordination requirements, making it an ideal test case for both TASP and SEW. When the permit enters the existing LaneClosure process, it would enter the Standardized Event Workflow (SEW) and be converted into a ROWDS event linked to the affected sidewalk segment, curb segment, and ADA ramp. The event would route review tasks to the responsible agencies through one shared record. Streets would review ROW occupancy, the ADA coordinator reviews accessible detour requirements, and PPA reviews curb or loading impacts if relevant.

After agency review and processing, the event would output in a standard format and input to the Trusted Asset-State Process (TASP). Because the event originates from an event workflow and has passed the required review steps, TASP would classify it as a high-confidence operational event. The system would automatically create a temporary asset state without overwriting the permanent record. When the approved time window ends, the temporary state would expire unless other events indicate that this event should remain active or trigger a permanent ROWDS asset update.

The pilot should be evaluated against a success matrix measuring reductions in off-system coordination, faster agency notification, improved event-to-asset linkage, and stronger evidence trails for future ROWDS updates.